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ALY 6050

Module 3 Project

Forecasting a Time Series

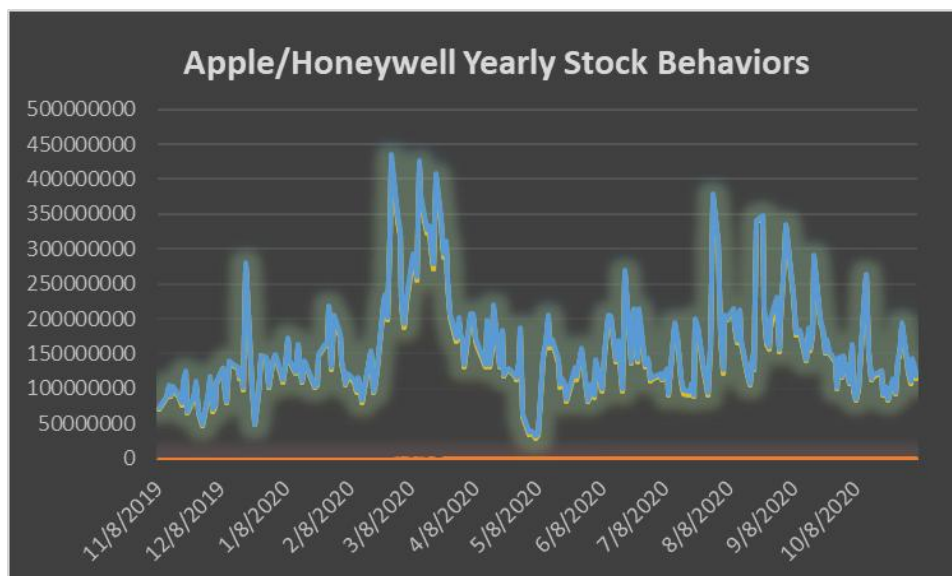
Introduction

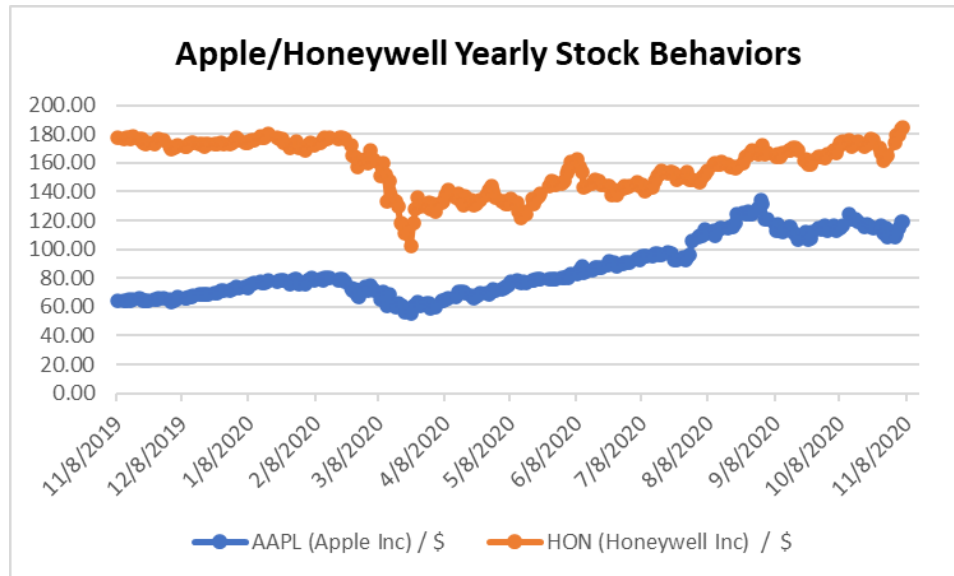
The module 3 project centers on two stocks called Apple and Honeywell International Inc and looks at the stock prices of each over the course of one calendar year that equals 252 days. The project has been divided into three parts which focus on short-term forecasting, long-term forecasting, and regression analysis. We will assess these stock prices by utilizing techniques learned this week that include exponential smoothing, 3-period weighted averages, and simple regression with plotting. After analysis of the three parts of the project a final question will address how much of the investment share from the stocks should be distributed to each stock which together equals 100 percent.

Analysis

Part 1: Short Term Forecasting

- (i) The Apple and Honeywell stocks are visually displayed in two 3D line graphs with the first plot showing the total volume of the stocks from each day. In the second plot the two stocks are separate from each other and show the value of each stock per day. The stocks in the first plot are the same in their trajectories by volume, and they remain consistent over time but have two spikes in the spring and then again in the fall of 2020. These parts of the year could be looked at historically to see if this is a common time of the year for volume to increase, or because this data was for the year 2020 that the pandemic might have made these behaviors an aberration due to those unforeseen circumstances. The second graph shows the prices of the stocks are constant until spring of 2020 when both tail off in price before gradually rising again for the rest of the year. This could relate to the volume increasing and be a pandemic factor due to when the stock prices fell before recovering.





- (ii) The values below show after the process of exponential smoothing of both stocks with successive values of 0.15, 0.35, 0.55, and 0.75 which lead to forecasting the stocks of day 253 which is indicated in bold. The first four values displayed are the Apple stocks and the four values on the right side are the Honeywell stocks.

113.15	110.81	110.03	109.93	170.86	172.33	174.91	177.26
113.39	112.19	112.63	113.55	172.07	174.63	177.11	178.50
114.20	114.51	116.04	117.51	173.75	177.66	180.50	182.08
114.88	115.97	117.50	118.39	175.33	179.97	182.58	183.72

With the formation of the values for each day of the stocks from exponential smoothing, the mean absolute percentage deviation can be calculated to see how accurate the forecasting of the stocks will be. The values below from the successive values used show low percentages between the values of the stocks and forecasting values which indicate that the forecast values are pretty accurate. While both stocks are close in their forecasting percentages, the Honeywell stock from each metric is lower than the Apple stock.

MAPD	MAPD	MAPD	MAPD	MAPD	MAPD	MAPD	MAPD
3.86	2.46	2.07	1.96	3.05	2.34	2.02	1.89

- (iii) The prices of each stock are forecasted with adjusted exponential smoothing by using the exponential forecast results of $\alpha=0.55$ from the previous calculation with new successive values. This is done by using values of alpha and trend parameters b which will provide the forecasts for stock 253. The four values shown on the left show the calculation using $\alpha=0.55$ while the right four values use the results of the first calculation with alpha to provide the forecasting values for stock 253.

Apple:

-0.16	0.07	0.80	2.21	112.47	112.70	113.42	114.83
0.38	0.91	1.97	3.23	116.41	116.94	118.01	119.26
0.54	1.05	1.74	1.73	118.04	118.54	119.24	119.22

Honeywell:

1.04	1.77	2.79	2.63	178.15	178.88	179.91	179.74
1.39	2.17	3.06	3.28	181.90	182.68	183.57	183.78
1.49	2.15	2.62	2.25	184.07	184.72	185.19	184.83

The four values in the diagram below are calculated using the mean absolute percentage error which is the accuracy of the forecasts relative to the actual values of each stock. Two of the four MAPE percentages from Apple are low while the other two are quite high, suggesting that there could be inconsistencies with the values or maybe there are some issues with overfitting. The values associated with the Honeywell stock are all low and could mean the accuracy is high, suggesting they are good at predicting trends and forecasting.

Apple:

	110.04	110.34	117.24	4.33
MAD				
	2.18	71.26	86.66	3.00

Honeywell:

	184.07	184.72	185.19	184.83
MAD				
	3.65	3.61	3.53	3.46

Part 2: Long Term Forecasting

- (i) The 3-period weighted averages are employed to show the direction of the stock prices based off past trends and patterns of the stock's prices. Increased weight in recent stocks is used with 50% weight being given to the most recent observation, followed by 30% weight to the observation before that, and 20% weight to the observation before that. The first 100 stock values are calculated to show the direction of each stock as indicated in the diagrams below.

Apple:

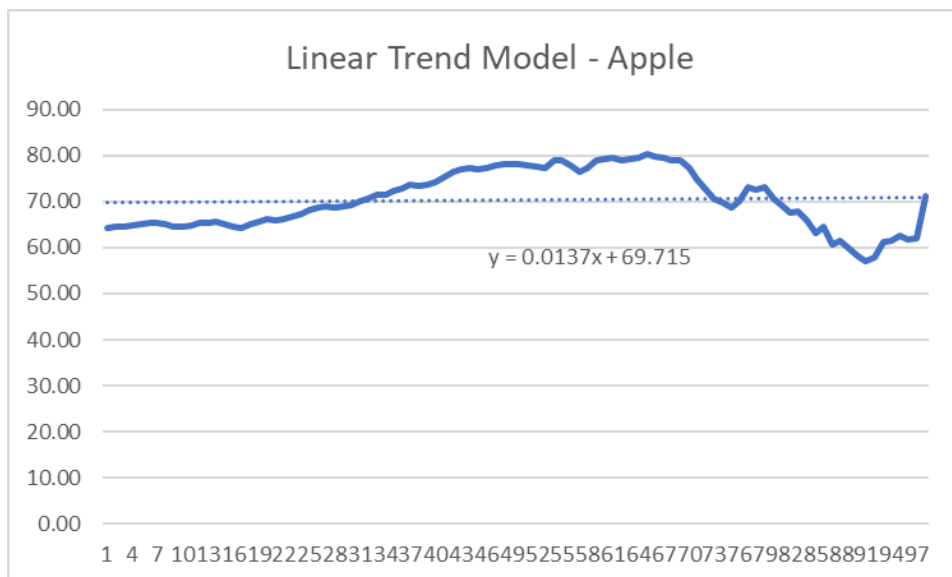
3 Period Weights	Apple Averages	3-Period Weight Averages
0.5	63.95	
0.30	64.46	
0.2	64.40	
	65.02	64.20
	64.57	64.55

Honeywell:

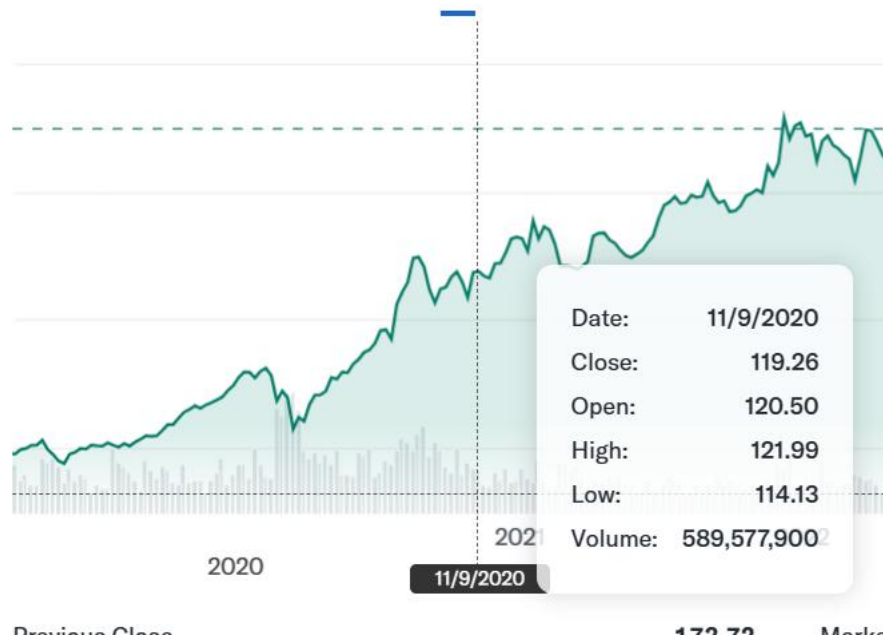
3 Period Weights	Honeywell Averages	3-Period Weighted Averages
0.5	177.03	
0.30	176.66	
0.2	177.81	
	177.75	177.07
	176.38	177.22

The stocks that range from 101 to 257 are evaluated using a line graph with an equation that was calculated based off the previous weighted averages. These prices will then determine the linear trend that will forecast the values of stocks 101 to 257. The Apple stocks recover after they decreased near the end of the first 100 days and end on day 257 with a price of \$73.24, almost \$10 more than the first day. The Honeywell stock seems to go in the opposite direction as it gradually goes down to \$77.41 on day 257, which is \$100 less than the first day. These final day prices from each stock are much lower than the actual prices of the stocks from that time period as indicated below. The Apple stock's trajectory could be going in that direction in the future, but it was not there yet.

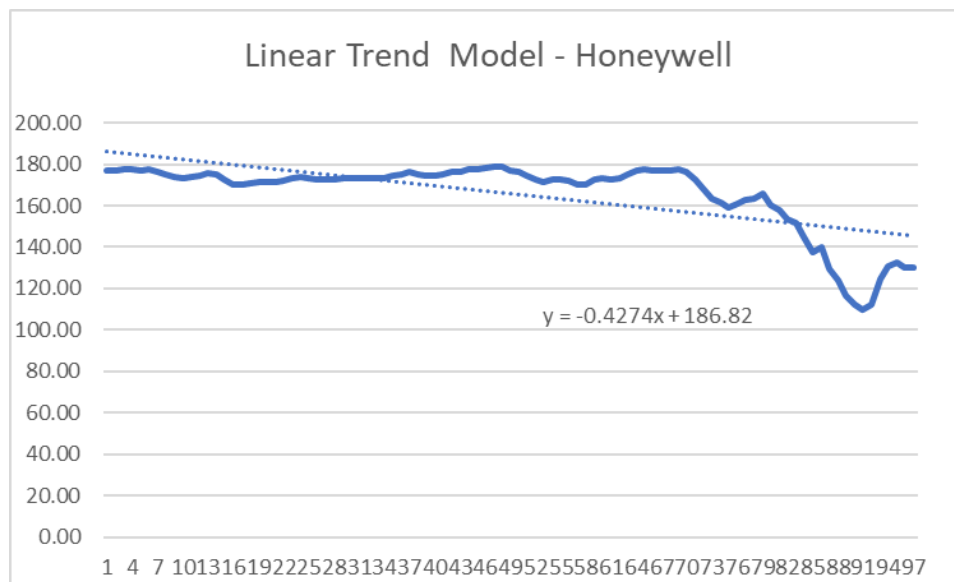
Apple:



118.82	73.15
118.69	73.17
	73.18
	73.20
	73.21
	73.22
	73.24



Honeywell:



183.28	79.97
184.27	79.54
	79.12
	78.69
	78.26
	77.83
	77.41



- (ii) The mean absolute percentage errors of each forecasted stock are calculated and then compared to the MAPE prices from part one of the project. Both stocks remain constant and have low values until around day 140 when both stocks MAPE's increase substantially. This contrasts with the MAPE's in part one where both stocks remain relatively constant for the duration of the calendar year. Several factors could be explored here with certain patterns and trends leading to increases in value or irregular events like the pandemic driving up these numbers.

Apple: Averages				Honeywell: Averages			
3-Period Weight	Averages	MAPE		3-Period Weighted	Averages	MAPE	
63.95				177.03			
64.46				176.66			
64.40				177.81			
65.02		64.20	0.82	177.75	177.07	0.68	
64.57		64.55	0.01	176.38	177.22	0.85	
114.75	73.14	41.61		178.91	80.40	98.51	
118.82	73.15	45.67		183.28	79.97	103.31	
118.69	73.17	45.52		184.27	79.54	104.73	
	73.18	73.18			79.12	79.12	
	73.20	73.20			78.69	78.69	
	73.21	73.21			78.26	78.26	
	73.22	73.22			77.83	77.83	
	73.24	73.24			77.41	77.41	

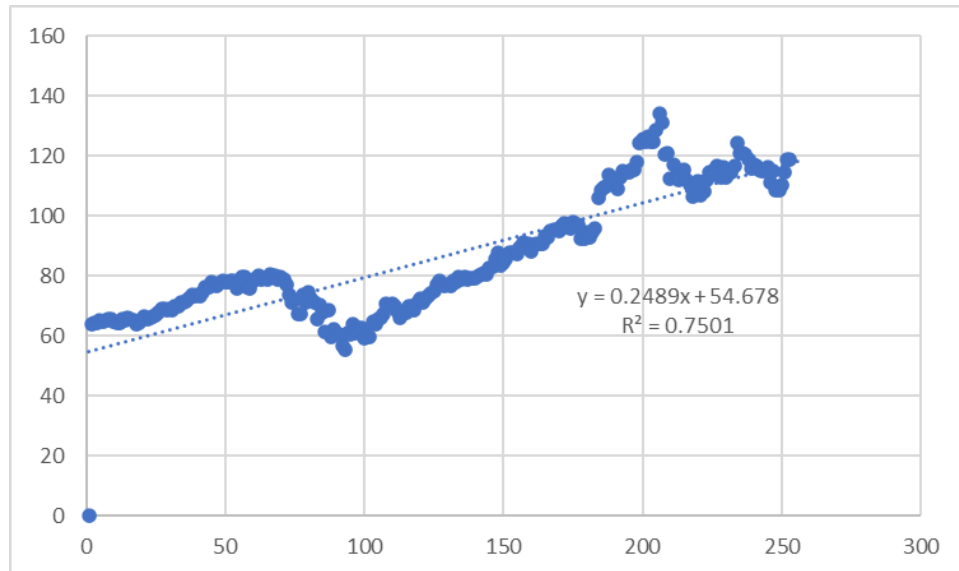
Part 3: Regression

- (i) Simple regression is incorporated into the stock price tables to predict the future values from day one to 257. The graph below shows the slope and intercept values used in the regression equation to calculate future stock price values. Both charts show a positive linear relationship between the variables time period and stock prices per day. After the predicted values were calculated the graphs below also indicate the same trends and patterns with both increasing, decreasing, and showing stability over the calendar year. This approach differs from the other formulas used in that there are no additional features added to the calculations. Regression analysis shows the correlation between the two

variables, leading to predicted values while not factoring in other measures such as recent trends and patterns or exponential smoothing of the values.

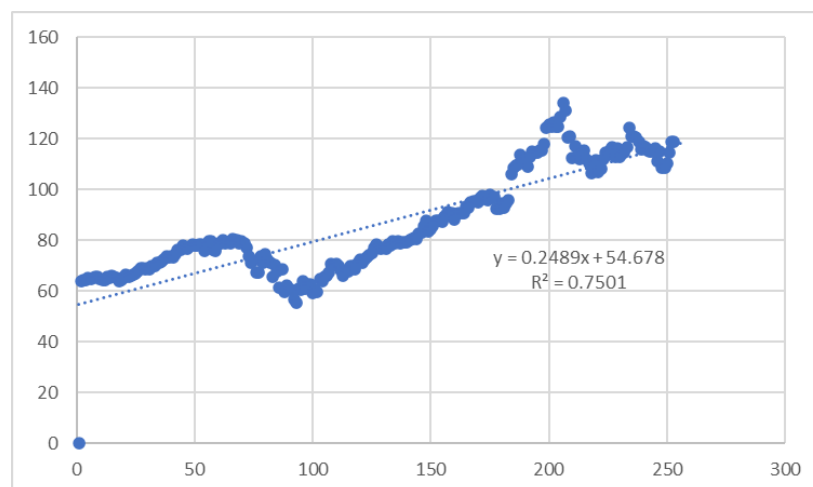
Apple:

Slope	0.24
Intercept	55.57
Correlation R	0.87
Determination R2	0.76



Honeywell:

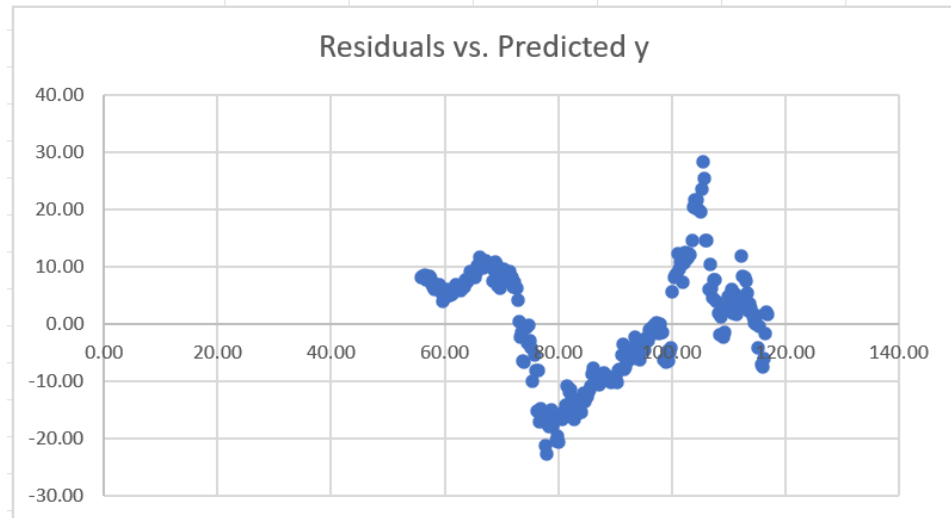
Slope	-0.03
Intercept	161.65
Correlation R	-0.13
Determination R2	0.02



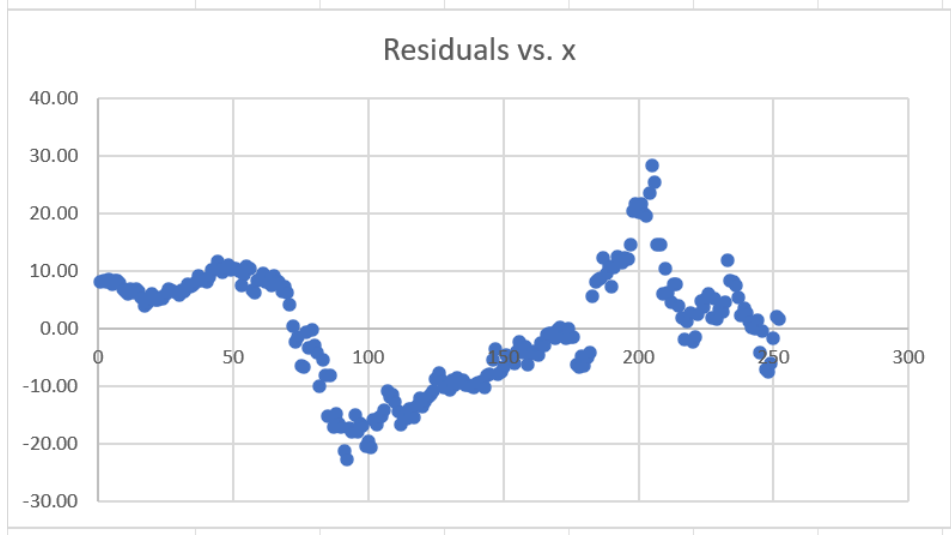
- (ii) A residual analysis of your simple regression to verify whether regression is appropriate to use for each of the given data.

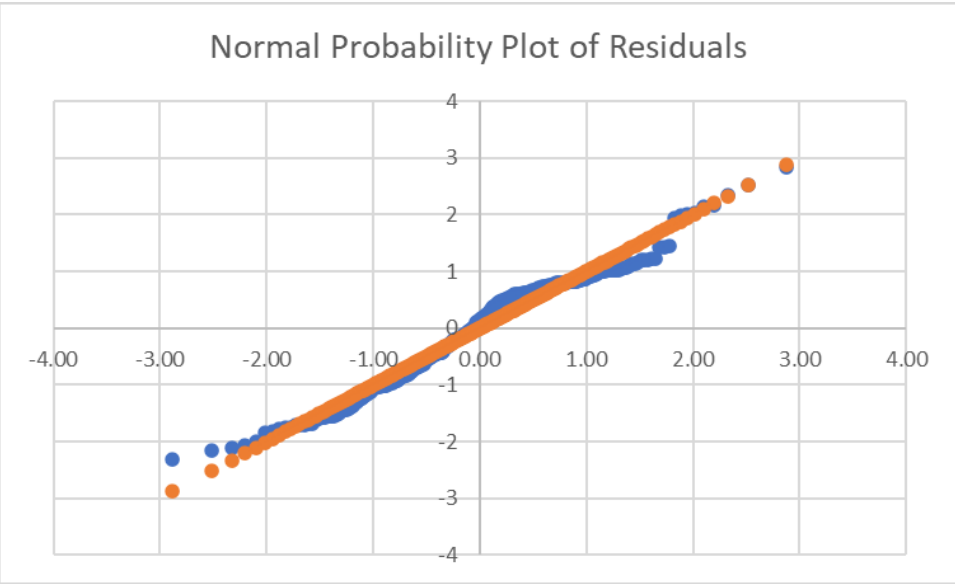
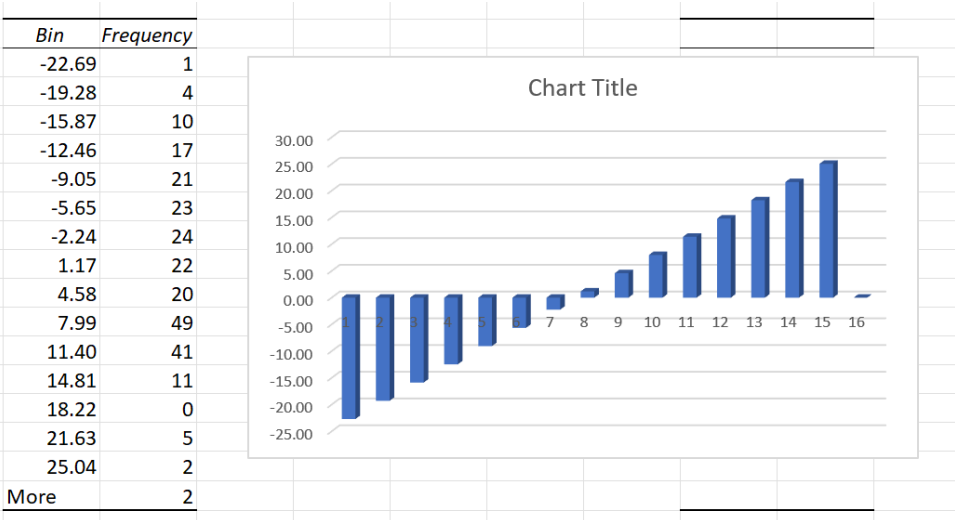
Apple:

Homoscedasticity



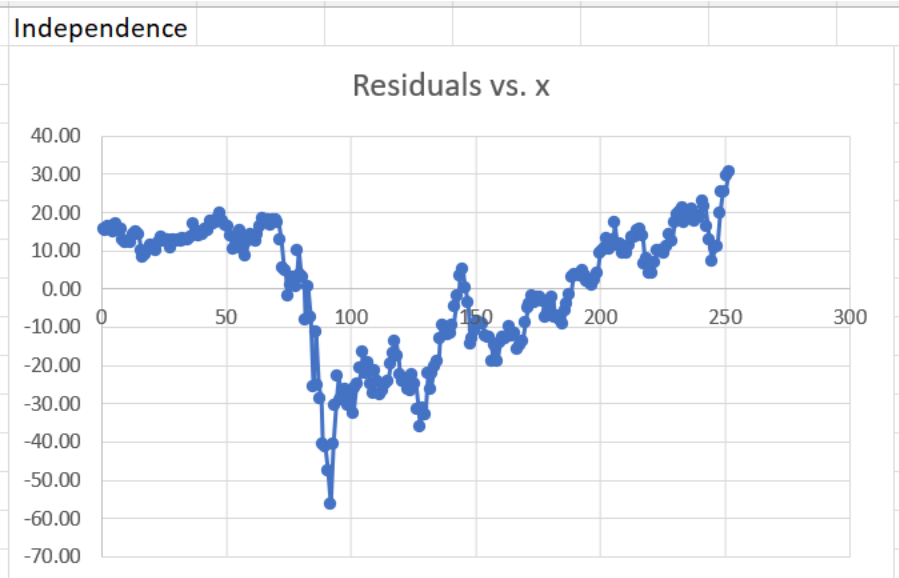
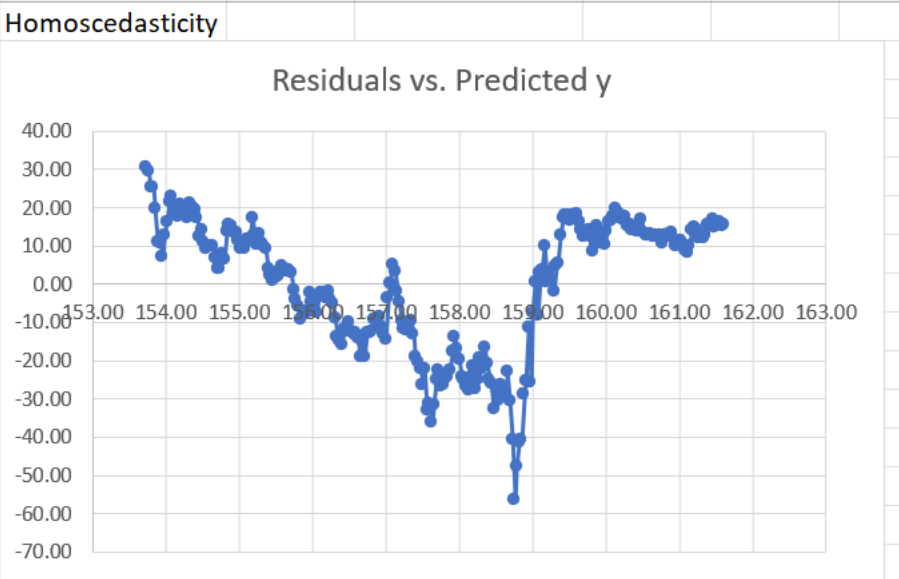
Independence

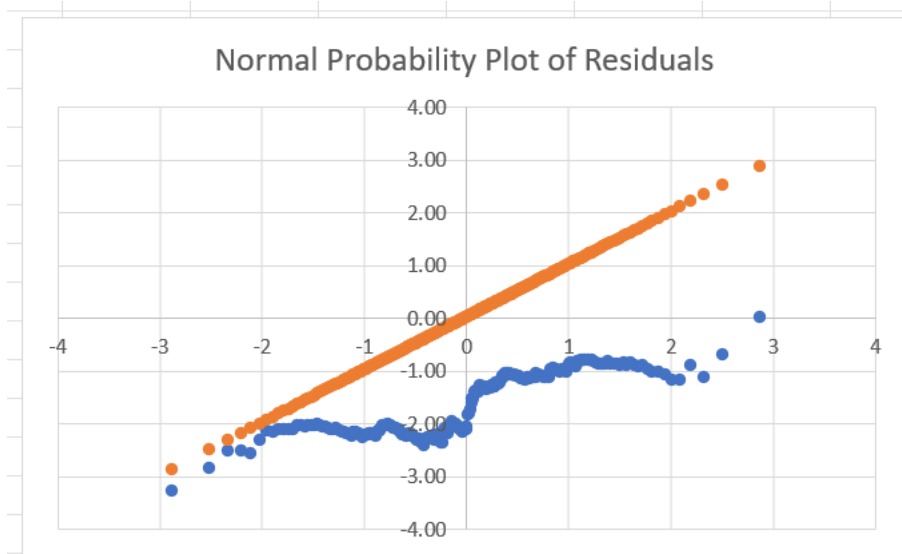
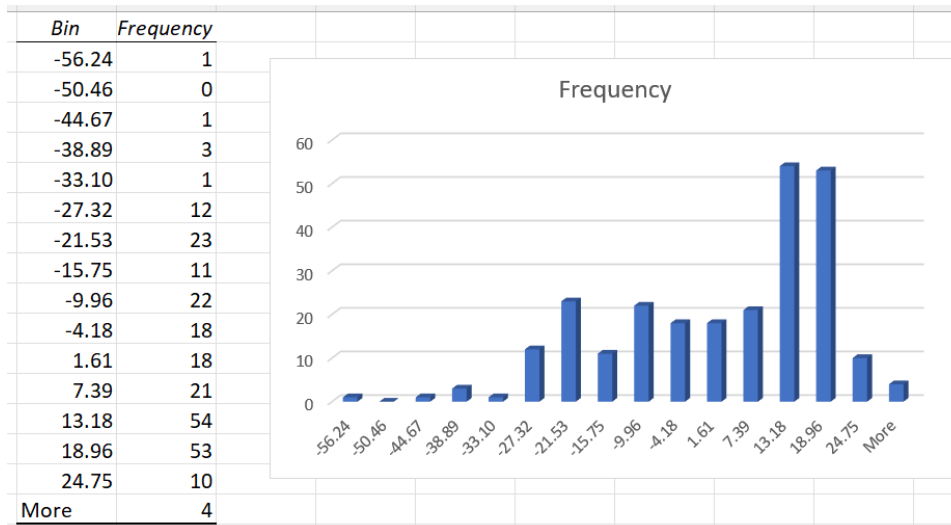




Chi-Squared Test						
		Bin	Frequency	Prob.	Exp. Freq.	Chi-Square Value
		-22.6897	1	0.0040	-2.31	-4.74 a=0.05
		-19.2808	4	0.0159	-1.96	-18.11 Df: 16-3 = 13
Residual Mean	0.25	-15.8719	10	0.0397	-1.62	-83.28 Critical value: 22.362
Residual Standard Deviation	9.94	-12.463	17	0.0675	-1.28	-261.24
		-9.05404	21	0.0833	-0.94	-514.08 There is substantial evidence
Ho: The Apple stock		-5.64513	23	0.0913	-0.59	-938.56 to reject the null hypothesis
follows a normal distribution		-2.23621	24	0.0952	-0.25	-2351.13 based on the chi square value
with a residual mean of 0.25 and		1.1727	22	0.0873	0.09	5170.10 of 6349.08 being much larger
a residual standard deviation		4.581613	20	0.0794	0.44	878.34 than the critical value of 22.362
of 9.94.		7.990527	49	0.1944	0.78	2986.02
		11.39944	41	0.1627	1.12	1417.77
Ha: The Apple stock does not		14.80835	11	0.0437	1.46	62.08
follows a normal distribution		18.21727	0	0.0000	1.81	1.81
with a residual mean of 0.25 and		21.62618	5	0.0198	2.15	3.78
a residual standard deviation		25.03509	2	0.0079	2.49	0.10
of 9.94.		More	2	0.0079	2.79	0.22
		Total:	252	1	4.18	6349.08

Honeywell:





Chi-Squared Test						
	Bin	Obs. Frequency	Prob.	Exp. Frequency	Chi-Square Value	
	-56.2417	1	0.004	0.004	248.00	a=0.05
	-50.4569	0	0	0.0016	0.00	Df: 16-3 = 13
Residual Mean: 0.00	-44.6721	1	0.004	0.004	248.00	Critical value: 22.362
Residual Standard Deviation: 17.05	-38.8873	3	0.012	0.0094	951.46	
	-33.1025	1	0.004	0.0256	37.09	There is substantial evidence
Ho: The Honeywell stock	-27.3177	12	0.048	0.0495	2885.14	to reject the null hypothesis
follows a normal distribution	-21.5329	23	0.091	0.0885	5931.49	based on the chi square value
with a residual mean of 0 and	-15.7481	11	0.044	0.1711	685.36	of 21376.75 being much larger
a residual standard deviation	-9.96325	22	0.087	0.2578	1833.68	than the critical value of 22.362
of 17.05.	-4.17844	18	0.071	0.4013	771.78	
	1.606366	18	0.071	0.9505	305.82	
Ha: The Honeywell stock does not	7.391173	21	0.083	0.6736	613.36	
follows a normal distribution	13.17598	54	0.214	0.7734	3663.14	
with a residual mean of 0 and	18.96079	53	0.21	0.8749	3105.53	
a residual standard deviation	24.7456	10	0.04	0.9394	87.39	
of 17.05.	More	4	0.016	0.9678	9.50	
	Total:	252	0.999	6.1924	21376.75	

Question: Suppose that you have decided to form a portfolio Π (P_i) consisting of the above two stock types (denote a share value of AAPL by X and that of HON by Y). You are, however, undecided as to what percentage of your investment should be allocated to the AAPL shares and

what percentage should be allocated to HON shares. Let these percentages be denoted by P and Q respectively (Obviously, $P + Q = 100\%$). In your opinion, what are good values to select for P and Q?:

P: Apple

Q: Longwell

While both stocks seem to show a lot of similarities in their behaviors and patterns over time, there are contrasts that stand out among each. Short term forecasting concludes that Honeywell has higher accuracies with its forecasting while the Apple stock has some forecasting values that suggest that there are potential issues. The long-term forecasting has Apple displaying more stability while Honeywell shows that over the year the stock prices are going in the wrong direction. Due to their correlations, I would look at the business model to see how goals would pertain to the risk of selecting either stock. Other factors to consider would be market outlooks and how each stock could be beneficial in the long term. The pandemic could have been a reason that both stocks that year were inconsistent in its performance, and with the potential risks with Honeywell in terms of its negative predictive values in long-term forecasting, I would put Apple at $P=0.55$ and Honeywell at $Q=0.45$. Over time this portfolio could be adjusted to where Longwell could show better long-term growth than Apple.

Conclusion

The project looked at several methods of projecting and forecasting prices of the stocks Apple and Honeywell and identifying from the results which stock would be more valuable and lucrative long term. During the calendar year that was assessed there were ebbs and flows for each stock that were based off several measures like exponential smoothing and residual analysis that presented short term results and future outlooks. These features can serve as a valuable tool for seeing how each stock will fare in the future.

References

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